Solution 1 Weyl Group of Type C_3

Problem 1 — Suppose Φ is of type C_3. Fix a base $\Delta = \{\alpha_1, \alpha_2, \alpha_3\}$ where α_3 is a long root. Write s_i for a simple reflection corresponding to α_i.

(1) Describe the Weyl group $\mathcal{W}_\Phi$ of Φ in terms of generators and relations.

(2) Write down 24 elements of $\mathcal{W}_\Phi$ in terms of s_i, $i = 1, 2, 3$.

(1) This is

$$\mathcal{W}_\Phi = \langle s_1, s_2, s_3 \mid s_i^2 = 1, (s_1 s_2)^3 = 1, (s_2 s_3)^4 = 1, (s_1 s_3)^2 = 1 \rangle$$

(2) The output of the following python code

```python
"""
Today, we will be explicitly enumerating the elements of the Weyl group
    of C3
"""

import numpy as np

SIMPLE = [
    np.array([0, 1, 1]),
    np.array([-1, -1, 0]),
    np.array([2, 0, 0]),
]


SIMPLE_REFLECTIONS = [np.identity(
    3) - 2 * np.outer(s, s) / np.dot(s, s) for s in SIMPLE]

WEYL_GROUP = [np.identity(3)]
WEYL_EXPANSION = [()]

DISC = 2 ** 16

WEYL_GROUP_SET = set([tuple(int(i) for i in w.reshape(9) * DISC + 0.5)
                      for w in WEYL_GROUP]) # for fast membership testing

i = 0
while i < len(WEYL_GROUP):
    for j, s in enumerate(SIMPLE_REFLECTIONS):
        test = np.matmul(s, WEYL_GROUP[i])
        test_tuple = tuple(int(i) for i in test.reshape(9) * DISC + 0.5)
        if test_tuple not in WEYL_GROUP_SET:
            WEYL_GROUP.append(test)
            WEYL_GROUP_SET.add(test_tuple)
            WEYL_EXPANSION.append(WEYL_EXPANSION[i] + (j,))
    i += 1

for s in WEYL_EXPANSION:
    res = " ".join(["\sigma_"+str(i+1) for i in s])
    print("&" + res + "\\\\")
```

```
print("The Weyl group of C3 has cardinality " + str(len(WEYL_GROUP)) +
    ".")
```

yields this result:

$$\mathbb{1}$$
$$\sigma_1$$
$$\sigma_2$$
$$\sigma_3$$
$$\sigma_1\sigma_2$$
$$\sigma_1\sigma_3$$
$$\sigma_2\sigma_1$$
$$\sigma_2\sigma_3$$
$$\sigma_3\sigma_2$$
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(observe that there are actually 48 elements in the Weyl group; this problem only asked for 24.)

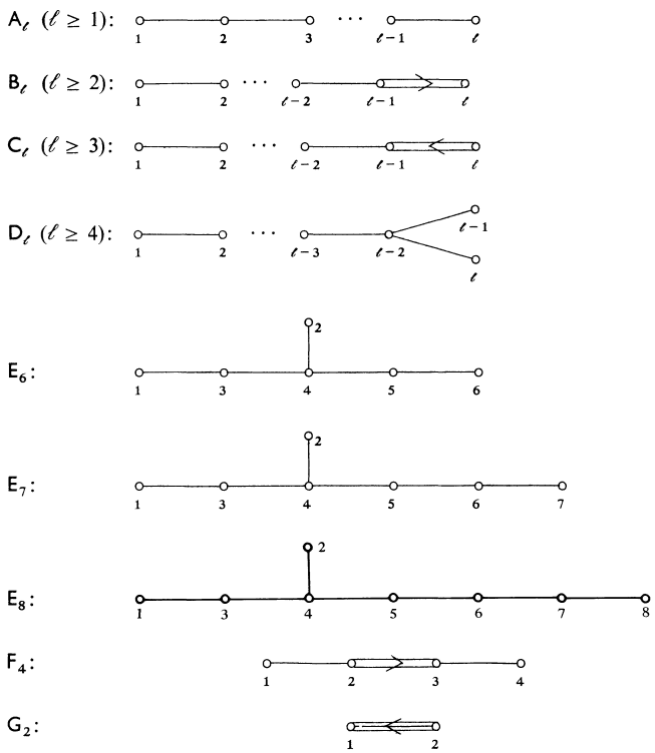
Solution 2 Properties of the Weyl Group

Problem 2 —

1. Show that if two vertices in a Dynkin diagram are connected by a single edge, then the corresponding simple roots are in the same $\mathcal{W}$-orbit.
2. Show that the Weyl group acts transitively on the set of all roots of the same length.
3. Let $w_0 \in \mathcal{W}$ is the element of maximal length. Show that $\ell(w_0) = \frac{1}{2}|\Phi|$ where $\ell(w_0)$ is the length of w_0.
4. For any $w \in \mathcal{W}$, show that $\ell(w_0 w) = \ell(w_0) - \ell(w)$.

1. If two vertices α and β are connected by a single edge, it implies that $1 = \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4\cos^2\theta$, where θ is the angle between α and β. Since $(\alpha, \beta) < 0$ (as they are both simple roots), we must have $\theta = \frac{2\pi}{3}$. Additionally, α and β must have the same length, as this is a geometric property. The inner products $\langle \alpha, \beta \rangle = -|\alpha|/|\beta|$ and $\langle \beta, \alpha \rangle = -|\beta|/|\alpha|$ must both be integers. Therefore, it is easy to see that $\sigma_\alpha \sigma_\beta \in \mathcal{W}$ sends α to β (and similarly, $\sigma_\beta \sigma_\alpha$ sends β to α).

2. Consider the classification of all irreducible root systems:



There exists a path of simple edges between any two roots of the same length. Thus, by (1), they must be in the same $\mathcal{W}$-orbit. We know that $\mathcal{W}$ acts transitively on any of its orbits, thus we conclude.

3. From Problem 1 on the previous pset, there is $w_0 \in \mathcal{W}$ which sends *all* of Φ^+ into Φ^-. It is well-known that for any $\sigma \in \mathcal{W}$, the length of σ is the number of positive roots which are sent to negative roots. So clearly this w_0 is of maximal length, and it's length is $|\Phi^+| = \frac{1}{2}|\Phi|$.

4. Any positive root $\alpha \in \Phi^+$ is sent into Φ^- by *exactly one of* w and $w_0 w$. Thus $\ell(w) + \ell(w_0 w) = |\Phi^+| = \ell(w_0)$, as desired.

Solution 3 Nilpotent and Borel Subalgebras

Problem 3 — Let $\mathfrak{g}$ be a semisimple Lie algebra. Let $\mathfrak{h}$ be a Cartan subalgebra of $\mathfrak{g}$. Let Φ be the set of $\mathfrak{h}$ roots with a base Δ.

1. Set $\mathfrak{n}_+ := \bigoplus_{\alpha \in \Phi_+} \mathfrak{g}_\alpha$. Prove that n_+ is nilpotent.
2. Set $\mathfrak{b} := \mathfrak{h} + n_+$. Prove that $\mathfrak{b}$ is a Borel subalgebra, that is, it is a maximal solvable Lie subalgebra of $\mathfrak{g}$.

1. Recall that for any two roots α, β we have $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$; basically, if you iterate this too many times, you'll get a root which is not in Φ_+, so the bracket will be zero.

   Specifically, let $\mathfrak{n}^{(1)} = \mathfrak{n}_+$ and $\mathfrak{n}^{(i+1)} = [\mathfrak{g}, \mathfrak{n}^{(i)}]$. It is clear that $\mathfrak{n}^{(i)} = \bigoplus_{c \in \Phi_+^n} \mathfrak{g}_c$ where Φ_+^n is the set of all sums of n positive roots. Since Φ is finite, there exists some N such that $\Phi_+^N = \emptyset$, so $\mathfrak{n}^{(N)} = \{0\}$ and $\mathfrak{n}_+$ is nilpotent.

2. This time, let $\mathfrak{b}^{(0)} = \mathfrak{b}$ and $\mathfrak{b}^{(i+1)} = [\mathfrak{b}^{(i)}, \mathfrak{b}^{(i)}]$.

   I claim $\mathfrak{b}^{(1)} = [\mathfrak{b}, \mathfrak{b}] \subset \mathfrak{n}_+$. This is completely obvious; $[\mathfrak{h} + \mathfrak{n}_+, \mathfrak{h} + \mathfrak{n}_+]$ contains $[\mathfrak{h}, \mathfrak{h}]$, which is zero, $[\mathfrak{h}, \mathfrak{n}_+]$, which is contained in $\mathfrak{n}_+$, and $[\mathfrak{n}_+, \mathfrak{n}_+]$, which is contained in $\mathfrak{n}_+$.

   Thus $\mathfrak{b}^{(1)} \subseteq \mathfrak{n}_+$ is nilpotent; then of course $\mathfrak{b}$ is solvable.

   To show $\mathfrak{b}$ is maximal, suppose for contradiction $\mathfrak{m}$ is a solvable subalgebra of $\mathfrak{g}$ containing $\mathfrak{b}$; then $\mathfrak{m}$ must contain some root space $\mathfrak{g}_{-\alpha}$ for some $\alpha \in \phi^+$. Then, $\mathfrak{m}$ contains g_α and $g_{-\alpha}$.

   This lets us pick elements $e \in \mathfrak{g}_\alpha, f \in \mathfrak{g}_{-\alpha}, h = [e, f]$ which span a $\mathfrak{sl}_2$ subalgebra of $\mathfrak{m}$.

   Since $\mathfrak{sl}_2$ is simple, $\mathfrak{m}$ cannot be solvable, and we're done.

Solution 4 Center of $U(\mathfrak{sl}_2)$

Problem 4 — Let $U(\mathfrak{sl}_2)$ be the universal enveloping algebra of $\mathfrak{sl}_2$ and e, h, f be the standard basis. Prove that

$$C = ef + fe + \frac{1}{2}h^2$$

is in the center of $U(\mathfrak{sl}_2)$.

We check the commutation relations.

$$\begin{aligned}
[C, e] &= [ef + fe + \frac{1}{2}h^2, e] \\
&= [f, e^2] + \frac{1}{2}(h[h, e] + [h, e]h) \\
&= [f, e]e + e[f, e] + eh + he \\
&= -he - eh + eh + he = 0 \\
[C, f] &= [ef + fe + \frac{1}{2}h^2, f] \\
&= [e, f^2] + \frac{1}{2}(h[h, f] + [h, f]h) \\
&= [e, f]f + f[e, f] - hf - fh \\
&= hf + fh - hf - fh = 0 \\
[C, h] &= [ef + fe + \frac{1}{2}h^2, h] \\
&= [e, h]f + e[f, h] + [f, h]e + f[e, h] \\
&= -2ef + 2ef + 2fe - 2fe = 0
\end{aligned}$$